

Influence of phosphate precursors on the structures of sodium aluminophosphate sols, gels and glasses

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Abstract

Transparent and homogeneous aluminophosphate gels and glasses have been widely synthesized through an aqueous sol–gel route, extending significantly the glass-forming range compared to that accessible via the melt-cooling route. Different phosphorus precursors, sodium polyphosphate (NaPO_3) and orthophosphate species (NaH_2PO_4 and/or H_3PO_4) were compared with regard to the macroscopic properties and the microscopic structure of the resultant gels and glasses as characterized by extensive high-resolution liquid- and solid-state NMR. Sodium polyphosphate solution results in a substantially wider composition range of homogenous gel formation than orthophosphate solutions, and the two routes produce significant structural differences in the sol and xerogel states. Nevertheless, the structures of the glasses obtained upon gel annealing above 400 °C are independent of the P-precursors used.

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1. Introduction

The sol–gel process has become an attractive technique for the preparation of oxide glasses with a great variety of chemical compositions [1–3]. The lower temperature processing and the mixing of the starting materials on a molecular scale offer unique opportunities of compositional and structural tailoring. Furthermore, the preparation of thin film is possible. The preparation of amorphous aluminophosphate powders and bulk glasses has attracted considerable attention in this regard [4–8], because these materials are useful in a number of applications as laser devices, catalysts and catalyst supports, solid-state batteries, and hermetic seals to materials with high thermal expansion coefficients [9–14]. Since commonly

used aluminum alkoxide precursors are highly reactive towards water, resulting in rapid hydrolysis and precipitation [15], the homogeneous introduction of alumina into glassy networks via the sol–gel route has been a great challenge [4–8]. Recently, a novel sol–gel synthesis based on aluminum lactate and sodium polyphosphate aqueous solution was developed in our laboratory, resulting in transparent gels and glasses along the compositional line $x\text{Al}_2\text{O}_3-(1-x)\text{NaPO}_3$ [8,16]. In the present study, we extend our previous work to formation of glasses within the entire ternary compositional diagram, and their structural characterization by extensive high-resolution liquid and solid-state NMR. In the present study, these results are used to explore and discuss the influence of the phosphorus precursor species on the structures of the sols, gels, and the final glasses formed upon gel annealing.

2. Experimental section

Sodium aluminophosphate gels and glasses were prepared from aqueous solutions within a pH range between

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2 and 4, using aluminum lactate (98%, Fluka), NaH_2PO_4 (99%, Fluka), NaPO_3 (95–97%, Fluka), sodium acetate (NaAc , 99%, Fluka) and H_3PO_4 (98%, Fluka) as precursors. H_3PO_4 (1 M) solution was prepared by dissolving solid H_3PO_4 into distilled water. NaPO_3 (2 M) solution was prepared by stirring sodium metaphosphate glass, which was obtained by cooling a melt of crystalline NaPO_3 from 700 °C to room temperature in a platinum crucible, in distilled H_2O for several hours in a glass container. The pH of the precursor solution was adjusted by adding a freshly prepared solution of nitric acid (1 M, puriss. P.a., Fluka) or ammonia (1 M, diluted from conc. ammonia solution, Aldrich) and controlled within 0.01 units by a pH meter (WTW pH320, Germany). In a typical sol preparation, 0.01 mol of aluminum lactate was dissolved into 20 ml distilled water, followed by the addition of required amount of NaH_2PO_4 (or NaPO_3). The appropriate amounts of H_3PO_4 and NaAc were used as supplemental additives to realize the targeted Na/P ratio of the product. After air-drying the above clear solutions for several days at temperatures ranging from ambient to 100 °C, transparent colorless xerogels were obtained. When using NaH_2PO_4 or H_3PO_4 as precursors, transparent aluminophosphate gel can be formed for $\text{P}/\text{Al} < 6$, whereas usage of sodium polyphosphate (NaPO_3) enables the formation of transparent gels over the composition range $\text{P}/\text{Al} \leq 10$.

Thermogravimetric analysis (TGA) of the xerogels, carried out on a NETZSCH STA409 instrument, reveal that volatile components (organics and water) are almost completely driven off below 400 °C. After heating the xerogels around 400 °C for several hours in air, homogeneous glasses are obtained. For those xerogels formed using mixed polyphosphate and orthophosphate precursor species, the heating rate for the conversion from gel to glass is generally required to be lower than 0.5–1 °C/min, or else phase separation and crystallization occurs. In order to eliminate entrapment of organics in the glass, the precursor gels need drying under vacuum ($p < 10^{-4}$ mbar) at 100–150 °C for 8 h before the subsequent conversion to glass

Table 1

Glass transition temperatures T_g of some representative sodium aluminophosphate glasses, prepared via both sol-gel (400–500 °C) and melt-cooling (1100–1500 °C) routes

Composition Na ₂ O– Al ₂ O ₃ –P ₂ O ₅	<i>T</i> _g (°C)		Melt-prepared glass
	Sol–gel glass		
	From NaH ₂ PO ₄ / H ₃ PO ₄	From NaPO ₃	
43.8–12.5–43.8	430	426	442
42.5–15–42.5	446	449	453
40–20–40	452	446	441
33.3–33.3–33.3	454	460	–
20–40–40	520	534	–
10–45–45	545	552	–
0–50–50	1060	–	–
10–40–50	525	NM	–
30–20–50	500	495	–
37.5–12.5–50	444	435	432
50–12.5–37.5	355	340	365
50–25–25	410	NM	–

NM – not measured.

is started. The glass compositions prepared in the present study are shown in Fig. 1. Note that the glass-forming range is significantly extended beyond that accessible by conventional melt-cooling, in particular, the alumina content can be increased considerably. The non-crystalline state of all of the samples studied was confirmed by the absence of any sharp X-ray power diffraction peaks (Guinier method, $\text{Cu K}\alpha_1$ radiation). In addition, the glass transition temperatures (T_g) were extracted from the DTA plots, and the values are summarized in Table 1.

^{27}Al and ^{31}P single resonance NMR measurements were recorded at resonance frequencies of 104.3 MHz (^{27}Al) and 160.4 MHz (^{31}P) on a Bruker DSX-400 spectrometer, using 4 mm magic angle spinning (MAS) NMR probes. Liquid-state NMR studies were conducted on stationary samples located inside zirconia rotors, whereas solid-state NMR data were obtained at MAS rotation frequencies of 8–12 kHz. Typical acquisition parameters were pulse length 5.0 μs (90° flip angle) for ^{31}P and 1.0 μs (30°, solid flip angle) for ^{27}Al , and recycle delays of 60 s (^{31}P) and 0.5 s (^{27}Al). The ^{27}Al and ^{31}P NMR chemical shifts are referenced to 1 M aluminum nitrate aqueous solution and 85% H_3PO_4 , respectively.

3. Results and discussion

3.1. Structure of the sols

To understand the effect of different phosphorus precursors (e.g. NaH_2PO_4 , H_3PO_4 and NaPO_3) on the structure of sols, gels and glasses, ^{27}Al and ^{31}P NMR experiments were recorded on the samples, prepared by using either orthophosphate (NaH_2PO_4 or H_3PO_4) or polyphosphate (NaPO_3) precursor solutions, or mixtures of both. Fig. 2 shows the ^{27}Al (left) and ^{31}P (right) liquid-state NMR spectra of $\text{Al}(\text{lact})_3/\text{NaPO}_3$ (1:2), $\text{Al}(\text{lact})_3/\text{NaH}_2\text{PO}_4/\text{NaPO}_3$ (1:1:1), $\text{Al}(\text{lact})_3/\text{NaH}_2\text{PO}_4$ (1:2), $\text{Al}(\text{lact})_3/\text{H}_3\text{PO}_4/\text{NaAc}$

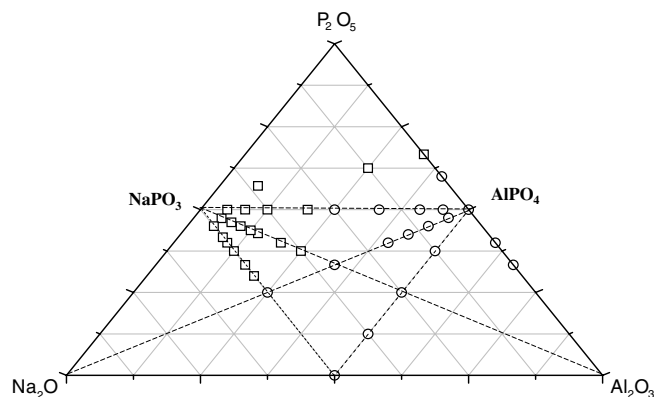


Fig. 1. Ternary phase diagram showing the compositions (mol%) of the glass samples prepared by the sol-gel route. Note that those glasses with $\text{P}/\text{Al} > 6$ can be prepared only using polyphosphate (NaPO_3) rather than orthophosphate precursors. o: Glasses which cannot be prepared by the conventional melt-quenching method.

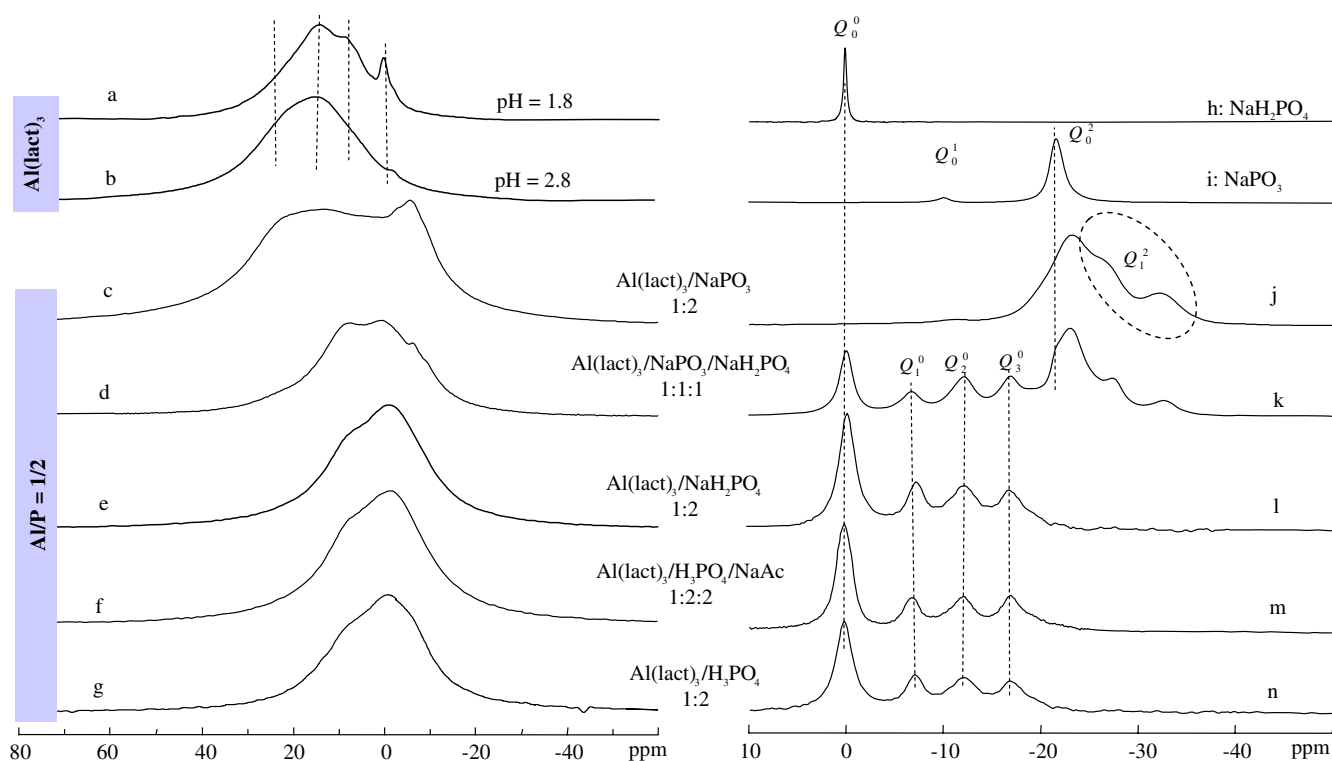


Fig. 2. ^{27}Al (left) and ^{31}P (right) liquid-state NMR spectra of $\text{Al}(\text{lact})_3/\text{NaPO}_3$ (1:2), $\text{Al}(\text{lact})_3/\text{NaH}_2\text{PO}_4/\text{NaPO}_3$ (1:1:1), $\text{Al}(\text{lact})_3/\text{NaH}_2\text{PO}_4$ (1:2), $\text{Al}(\text{lact})_3/\text{H}_3\text{PO}_4/\text{NaAc}$ (1:2:2), and $\text{Al}(\text{lact})_3/\text{H}_3\text{PO}_4$ (1:2) solutions, at pH value around 2.8.

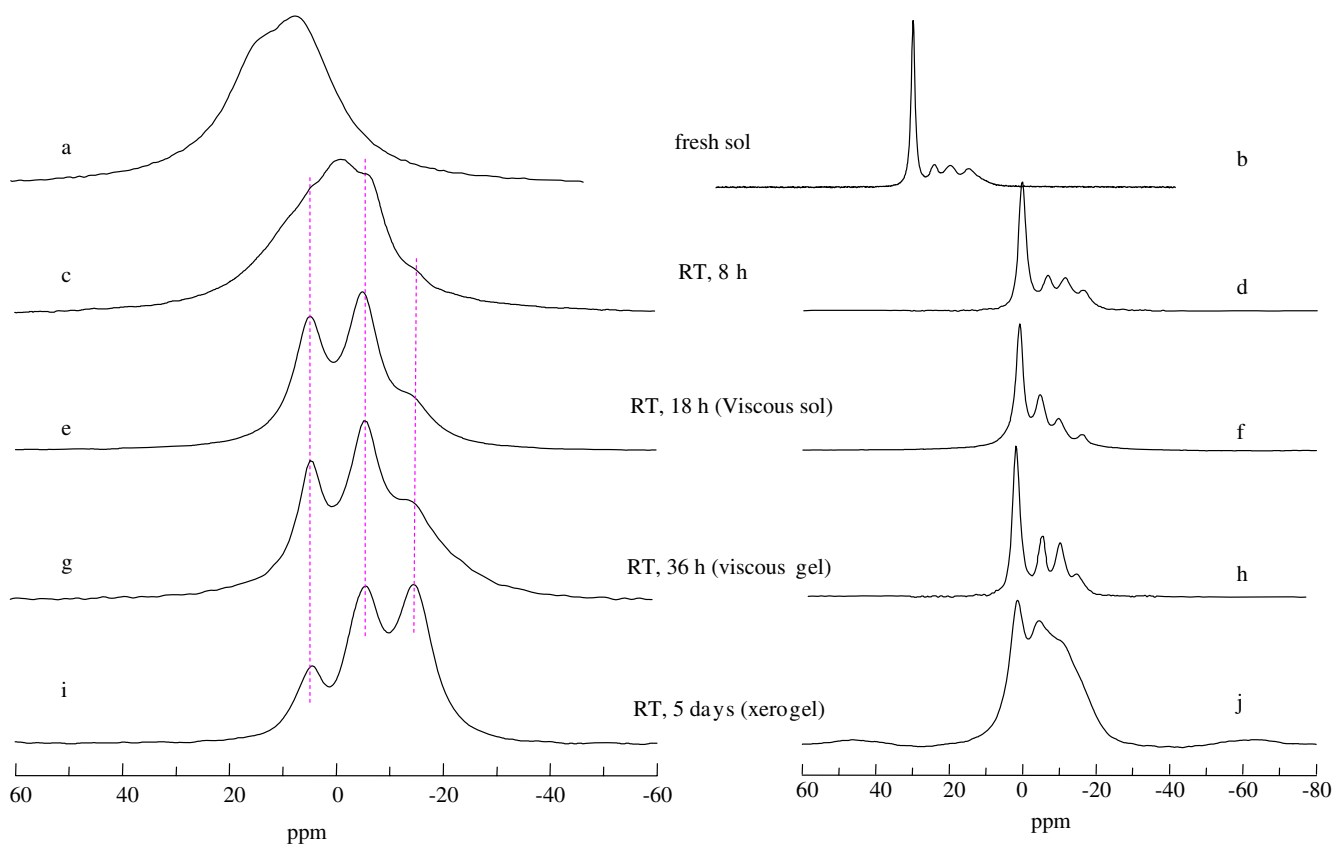
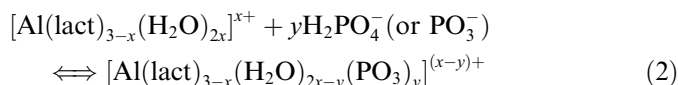
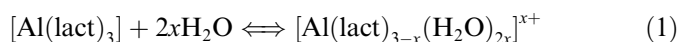


Fig. 3. ^{27}Al (left) and ^{31}P (right) NMR spectra of $\text{Al}(\text{lact})/\text{NaH}_2\text{PO}_4$ (1:3) samples, subjected to air-drying as a function of time from sol to xerogel conversion. (a–h) measured under static conditions; (i, j) measured under MAS conditions.

(1:2:2), and $\text{Al}(\text{lact})_3/\text{H}_3\text{PO}_4$ (1:2) solutions, at a pH value around 2.8. First of all, both the ^{27}Al (Fig. 2-left) and ^{31}P (Fig. 2-right) spectra indicate that the sodium source (either NaH_2PO_4 or NaAc) influences neither the local environments of the aluminum nor those of the phosphorus atoms in solution. Addition of phosphorus precursors (NaH_2PO_4 , H_3PO_4 or NaPO_3) into $\text{Al}(\text{lact})_3$ solution (Fig. 2(a) and (b)) produces a pronounced lower-frequency shift in the ^{27}Al signals, reflecting aluminum ligation to phosphate as described by equilibria of the general type



involving multiple combinations of x and y . Here, the phosphate ligands are only formally represented as PO_3^- species. Thus, the complicated ^{27}Al NMR spectra can be understood in terms of four signal components, $[\text{Al}(\text{lact})_3]$ (near 22 ppm), $[\text{Al}(\text{lact})_2(\text{H}_2\text{O})_{2-y}(\text{PO}_3)_y]^{2-y}$ (near 12 ppm), $[\text{Al}(\text{lact})_1(\text{H}_2\text{O})_{4-y}(\text{PO}_3)_y]^{1-y}$ (near 5 ppm) and $[\text{Al}(\text{H}_2\text{O})_{6-y}(\text{PO}_3)_y]^{3-y}$ (from 0 to –10 ppm) [16,17]. As shown in Fig. 2(c)–(g), with the substitution of NaPO_3 by NaH_2PO_4 (or $\text{H}_3\text{PO}_4/\text{NaAc}$) the ^{27}Al signal pattern shifts successively towards lower frequencies, reflecting

the fact that dissolved $\text{Al}(\text{lact})_3$ reacts with the orthophosphate species (NaH_2PO_4 or H_3PO_4) more efficiently than with dissolved polyphosphate species (NaPO_3) in solution.

The lower extent of P–O–Al connectivity in the $\text{NaPO}_3/\text{Al}(\text{lact})_3$ solutions also becomes evident in the corresponding ^{31}P NMR spectra. As illustrated in Fig. 2-right of these solutions, the dominant species in $\text{Al}(\text{lact})_3/\text{NaPO}_3$ solutions (Fig. 2(j)) are mainly polyphosphate (Q^2) species bound to zero (Q_0^2 , –21 ppm) and one (Q_1^2 , –24, –28 and –34 ppm) aluminum neighbors. In contrast, the ^{31}P NMR spectra of $\text{Al}(\text{lact})_3/\text{orthophosphate}$ (Fig. 2(l)–(n)) consist of four peaks around 0, –7, –12 and –17 ppm, which are assigned to orthophosphate (Q^0) species bound to zero (Q_0^0), one (Q_1^0), two (Q_2^0) and three (Q_3^0) aluminum atoms, respectively [17]. Solutions containing both types of P-precursor species (i.e. $\text{Al}(\text{lact})_3/\text{NaH}_2\text{PO}_4/\text{NaPO}_3$ (1:1:1) solutions (Fig. 2(k))) can be understood in terms of a superposition of those species present in each individual type of precursor solution.

3.2. Structural evolution from sol to gel

Upon air-drying, transparent colorless sodium aluminophosphate xerogels are formed. Fig. 3 shows the ^{27}Al (left) and the ^{31}P (right) NMR spectra of the $\text{Al}(\text{lact})_3/\text{NaPO}_3$

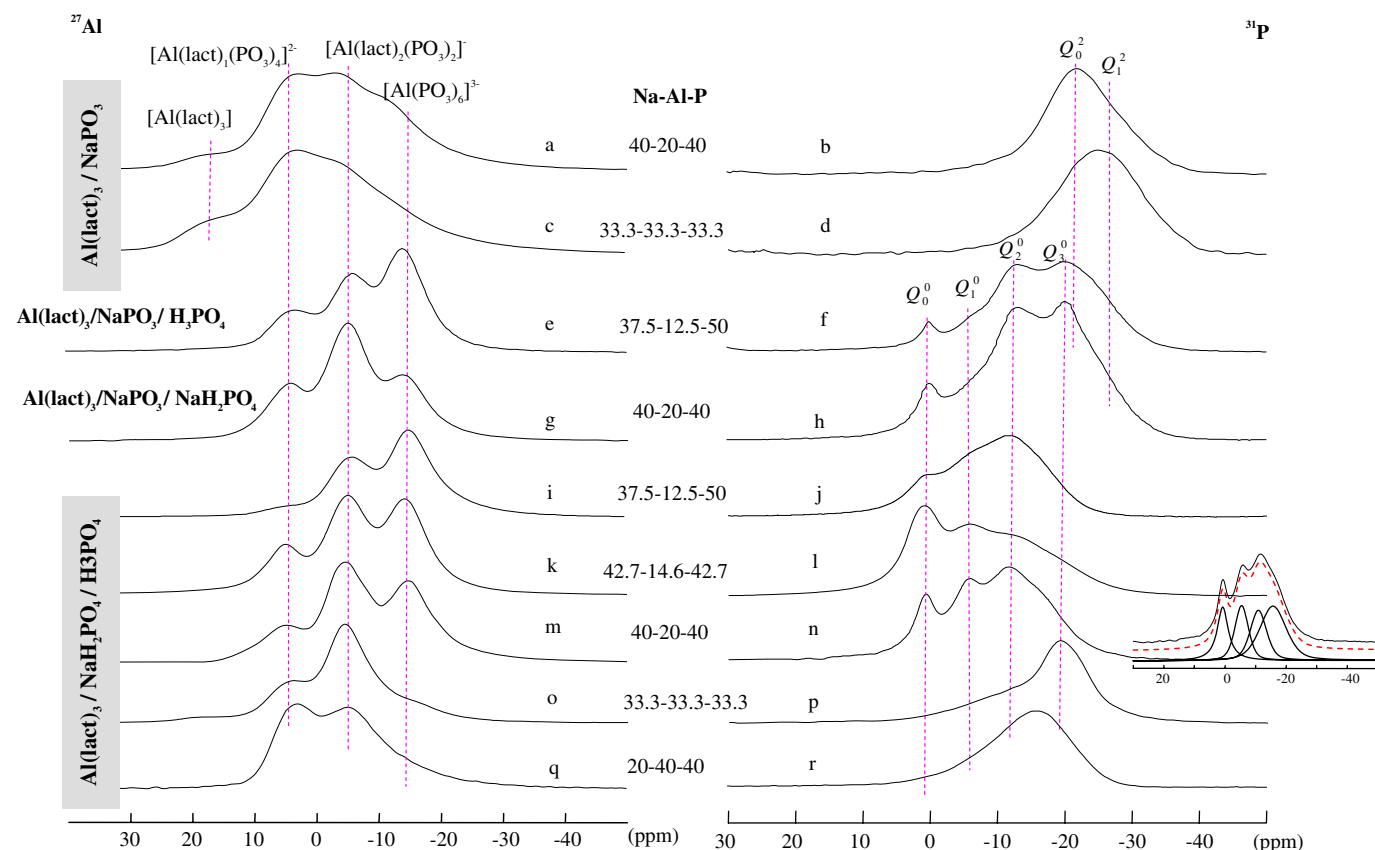
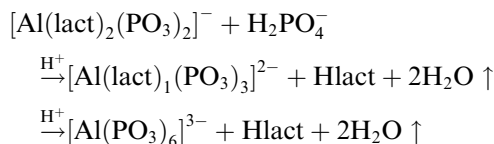


Fig. 4. ^{27}Al (left) and ^{31}P (right) MAS NMR spectra of some representative xerogels. All xerogels are prepared at 50 °C from corresponding precursor sols with pH around 3.0. The phosphorus precursors used in these samples are: (a–d) polyphosphate (NaPO_3) species; (e–h) mixtures of polyphosphate (NaPO_3) and orthophosphate (NaH_2PO_4 or H_3PO_4) species; (i–r) orthophosphate (NaH_2PO_4 and H_3PO_4) species. The inset shows the suggested spectral deconvolution of trace n into components due to the Q_m^0 ($0 \leq m \leq 3$) species.

$\text{NaH}_2\text{PO}_4 = 1:3$ (pH 3.2) samples as a function of drying-time at ambient conditions, reflecting the successive conversion from sol to gel. First of all, with increasing air-drying time the ^{27}Al NMR peak intensity distribution shifts towards lower-frequency, reflecting an increase in the average P/Al connectivity. This is consistent with the results from the ^{31}P spectra, which indicate that the concentrations of Al-bound phosphorus species (i.e. Q_1^0 , Q_2^0 and Q_3^0) increase with increasing drying time. In the viscous sols observed after drying for 18 h (Fig. 3(e)), the ^{27}Al NMR spectra are dominated by three well-separated resonances at 4, -5 , and -14 ppm, respectively. These signals can be assigned to $[\text{Al}(\text{lact})_2(\text{PO}_3)_2]^-$, $[\text{Al}(\text{lact})_1(\text{PO}_3)_4]^{2-}$ and $[\text{Al}(\text{PO}_3)_6]^{3-}$ species [16,17], as confirmed previously by $^{27}\text{Al}\{^{31}\text{P}\}$ REDOR data measured on the corresponding xerogel samples. Subsequent air-drying results in further spectral changes (see Fig. 3(g)–(i)), suggesting that direct acid–base reactions of the type



promote polymerization and thereby increase the extent of P/Al connectivity.

Fig. 4-left summarizes typical ^{27}Al (left) and ^{31}P (right) MAS NMR spectra of some representative xerogels formed using polyphosphate (NaPO_3 , Fig. 4(a)–(d)), orthophosphate (NaH_2PO_4 and H_3PO_4 , Fig. 4(i)–(r)) or both (Fig. 4(e)–(h)) as precursors. The ^{27}Al spectra show the four signal components identified above. Their intensity distribution reveals – for each composition – on average more Al–O–P linkages in the xerogel if NaH_2PO_4 instead of NaPO_3 was used in the preparation, in agreement with the situation in the corresponding sol (Fig. 2). Likewise, the ^{31}P NMR spectra of these xerogels (Fig. 4-right) can be easily understood based on the corresponding solution-state work. In the xerogels formed from NaPO_3 (Fig. 4(b) and (d)) as the P-precursor, the ^{31}P NMR spectra are still dominated by Q_2^0 (-21 ppm) and Q_1^2 (-25 ppm) species, and only a minor concentration of Q_1^1 units is detectable at -12 ppm. In contrast, the ^{31}P spectra (Fig. 4(j)–(r)) of xerogel formed using orthophosphate precursors (NaH_2PO_4 and H_3PO_4) reveal the Q_m^0 ($0 \leq m \leq 3$) units as the only species present. Finally, the xerogels formed using mixed polyphosphate and orthophosphate precursor solutions (Fig. 4(f) and (h)) contain all of the species present in each type of the above xerogels.

When using orthophosphate (NaH_2PO_4 and H_3PO_4) as precursors, transparent aluminophosphate gel can be formed only at $\text{P}/\text{Al} < 6$. For solutions and gels with larger

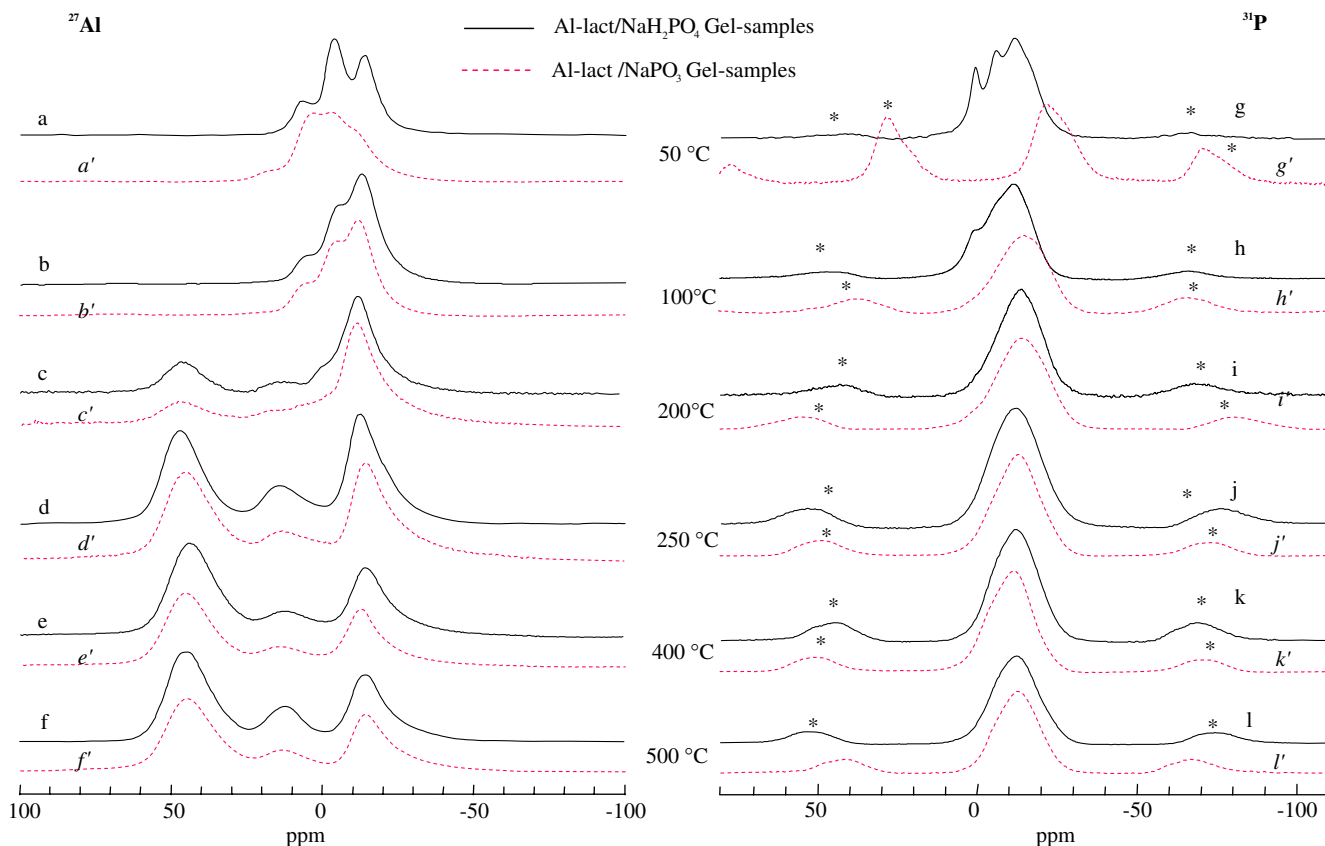


Fig. 5. ^{27}Al (left) and ^{31}P (right) MAS NMR spectra of $40\text{Na}_2\text{O}-20\text{Al}_2\text{O}_3-40\text{P}_2\text{O}_5$ gel-samples obtained from $\text{Al}(\text{lact})_3/\text{NaH}_2\text{PO}_4$ and $\text{Al}(\text{lact})_3/\text{NaPO}_3$ sols, subjected to different processing temperatures during gel to glass conversion. Spinning sidebands are indicated with asterisks. (a–l) $\text{Al}(\text{lact})_3/\text{NaH}_2\text{PO}_4$ series (solid curve); (a'–l') $\text{Al}(\text{lact})_3/\text{NaPO}_3$ series (dashed curve).

P/Al ratios, the ^{31}P NMR spectra (not shown here) reveal the dominance of Q_0^0 species, which contribute nothing to network connectivity. As a consequence, polymerization and gelation are suppressed, and drying results in microcrystalline powders instead. In contrast, the polyphosphate linkages in $\text{Al}(\text{lact})_3/\text{NaPO}_3$ sols generate more favorable conditions for gelation at large P/Al ratios.

3.3. Structural evolution from gel to glass

As stated above, the structures of the xerogels formed using orthophosphate and polyphosphate as precursors exhibit significant differences. To monitor and compare their respective structural rearrangements during the gel-to-glass transformation, ^{27}Al and ^{31}P MAS NMR spectra were recorded on the gel samples taken to different heating steps. Fig. 5 summarizes the results for Na/Al/P 40:20:40 samples prepared using either NaH_2PO_4 (solid curve) or NaPO_3 (dashed curve) precursor. The overall ^{27}Al spectral evolutions (Fig. 5-left) for both $\text{Al}(\text{lact})_3/\text{NaH}_2\text{PO}_4$ (solid curve) and $\text{Al}(\text{lact})_3/\text{NaPO}_3$ (dashed curve) series are quite similar (Fig. 5-left), in spite of the different fractional concentrations of each type of $[\text{Al}(\text{lact})_{3-x}(\text{PO}_3)_{2x}]^{x-}$ site in the initial xerogels (Fig. 5(a) and (a')): with increasing the gel-processing temperatures (Fig. 5(a)–(c) and (a')–

(c')), lactate moieties linked to aluminum atoms are successively replaced by phosphate species, resulting in an increased number of P–O–Al linkages. Gels annealed above 200 °C (Fig. 5(c) and (c')) show two new ^{27}Al resonances near 40 and 10 ppm attributable to $\text{Al}(\text{OP})_4$ and $\text{Al}(\text{OP})_5$ sites [18–20], respectively, indicating a major structural rearrangement in the network of the gels or route to the glassy state. The concentrations of these units increase with increasing annealing temperature. Calcination at 400 °C for several hours results in the final aluminum site distribution, which is the same for both gel-derived glasses (Fig. 5(e) and (f) and (e') and (f')).

A similar convergence is seen in the ^{31}P spectra. While the xerogels processed at 50 °C and 100 °C are dramatically different between the $\text{Al}(\text{lact})_3/\text{NaH}_2\text{PO}_4$ and $\text{Al}(\text{lact})_3/\text{NaPO}_3$ samples, heating to 200 °C and above results in very similar structural evolutions in both preparations. Based on the ^{31}P chemical shifts, the dominant phosphorus environments in the final glasses are Q_2^0 , Q_1^1 and Q_2^1 units, regardless of the phosphorus precursor used. A closer correlation of the ^{31}P and ^{27}Al spectra seen for both sample preparations reveals which of the chemical rearrangement processes are directly associated with each other. Thus the $Q^2 \rightarrow Q^1$ depolymerization observed at 100 °C in the $\text{Al}(\text{lact})_3/\text{NaPO}_3$ -based gels

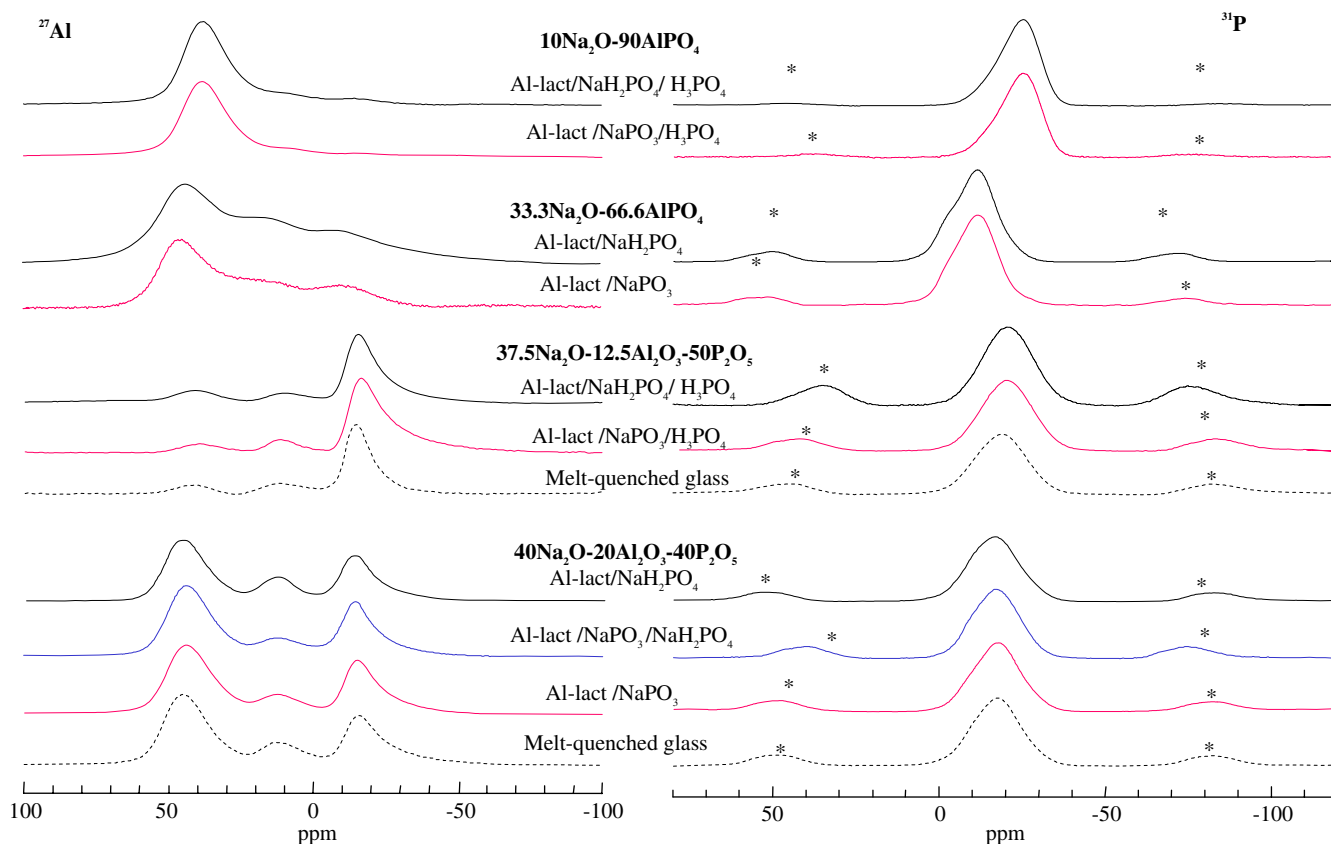


Fig. 6. ^{27}Al (left) and ^{31}P (right) MAS NMR spectra of some representative sodium aluminophosphate glasses, annealed at 400 °C. These spectra indicate that there is virtually no structural difference between glasses prepared from either orthophosphate or polyphosphate (NaPO_3) precursors. The dashed traces represent the spectra of the corresponding melt-prepared glasses, where are available.

seems to be directly accompanied by the elimination of lactate from the aluminum coordination environment, resulting in a significant increase of Al/P connectivity at this particular stage. In contrast, no further changes in the phosphorus coordination environments occur as the AlO_4 and AlO_5 units are formed during the gel $\rightarrow$ -glass transition at higher annealing temperatures. The situation is different in the $\text{Al}(\text{lact})_3/\text{NaH}_2\text{PO}_4$ xerogels, the structure of which remains essentially unaffected by sample-processing at 100 °C. For these gels, the major structural change occurs at the 200 °C heating step where $Q_m^0 \rightarrow Q_m^1$ repolymerization of the phosphorus species occurs, concomitant with the formation of the $\text{Al}(\text{OP})_4$ and $\text{Al}(\text{OP})_5$ units associated with the gel-to-glass transition. Higher annealing temperatures lead to a further consolidation of the glass structure formed, producing only minor structural changes and no further significant differences in both preparations.

4. Conclusions

Compared with the conventional melt-quenching method in which the alumina content of sodium aluminophosphate glasses is generally limited to less than 25 mol% [21,22], the sol–gel route described here affords glasses with much higher alumina concentrations (see Fig. 1). As illustrated in Figs. 2 and 4, the P-precursor species (polyphosphate or orthophosphate) used in their sol–gel preparations influence greatly the structures of the sols and gels, but the structures of glasses obtained from annealing these gels around 400 °C are virtually the same. Fig. 6 illustrates that this conclusion is true for glasses with very different Na/Al/P ratios within the homogeneous glass-forming range in this system. Note that the NMR spectra (Fig. 6) reveal a strong dependence of glass structure on composition.

Acknowledgements

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References

- [1] T. Woignier, J. Phalippou, J. Zarzycki, *J. Non-Cryst. Solids* 63 (1984) 117.
- [2] C.J. Brinker, G.W. Scherer, *Sol–Gel Science: The Physics and Chemistry of Sol–Gel Processing*, Academic, San Diego, 1990.
- [3] L.L. Hench, J.K. West, *Chem. Rev.* 90 (1990) 33.
- [4] L. Montage, G. Palavit, M. Draoui, *J. Non-Cryst. Solids* 155 (1993) 115.
- [5] E.C.O. Lima, F. Galembeck, *J. Colloid Interf. Sci.* 166 (1994) 309.
- [6] C.G. Lugmair, T.D. Tilley, A.L. Rheingold, *Chem. Mater.* 11 (1999) 1615.
- [7] M.A. Harmer, A.J. Vega, R.B. Flippen, *Chem. Mater.* 6 (1994) 1903; S. Prabakar, K.J. Rao, C.N.R. Rao, *Mater. Res. Bull.* 26 (1991) 805.
- [8] L. Zhang, J.C.C. Chan, H. Eckert, G. Hensch, L. Hoyer, G.H. Frischat, *Chem. Mater.* 15 (2003) 2702.
- [9] M.J. Weber, *J. Non-Cryst. Solids* 123 (1990) 208; L. Zhang, H. Hu, C. Qi, F. Li, *Opt. Mater.* 17 (2001) 371.
- [10] S.W. Martin, *J. Am. Ceram. Soc.* 74 (1991) 1767.
- [11] M.J.G. Jak, E.M. Kelder, Z.A. Kaszkur, J. Pielaszek, Schoonman, *Solid State Ionics* 119 (1999) 159.
- [12] J.A. Wilder, *J. Non-Cryst. Solids* 38&39 (1980) 879.
- [13] G.J. Hutchings, I.D. Hudson, D.G. Timms, *J. Chem. Soc., Chem. Commun.* (1994) 2717.
- [14] G.J. Hutchings, I.D. Hudson, D. Bethell, D.G. Timms, *J. Catal.* 188 (1999) 291.
- [15] B.E. Yoldas, *Am. Ceram. Soc. Bull.* 54 (1975) 289.
- [16] L. Zhang, H. Eckert, *Solid State Nucl. Magn. Reson.* 26 (2004) 132.
- [17] L. Zhang, H. Eckert, *J. Mater. Chem.* 14 (2004) 1605.
- [18] R.K. Brow, R.J. Kirkpatrick, G.L. Turner, *J. Am. Ceram. Soc.* 76 (1993) 919.
- [19] D.P. Lang, T.M. Alam, D.N. Bencoe, *Chem. Mater.* 13 (2001) 420, references therein.
- [20] J.M. Egan, R.M. Wenslow, K.T. Mueller, *J. Non-Cryst. Solids* 261 (2000) 115.
- [21] R.K. Brow, *J. Am. Ceram. Soc.* 76 (1993) 913.
- [22] H.N. Ng, C. Calvo, *Can. J. Chem.* 51 (1973) 2613.